

PAU MONTEROSA

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CONTACT

- **Location:** Barcelona
- **GitHub:** PauMonterosa
- **Web:** paumonterosa.github.io

SKILLS

Programming

- Python
- MATLAB / Simulink
- C++ / SQL

Tools

- $\text{\LaTeX}$ / TikZ
- Git / Docker
- LabVIEW
- Excel

LANGUAGES

- **English:** C1-Advanced
- **Spanish:** Native
- **Catalan:** Native
- **French:** B1

EDUCATION

- B.Sc. in Engineering Physics**

2023 – 2026

Universitat Politècnica de Catalunya
 - Focus: Biophysics & Signal Processing.
 - Relevant project: "TakeDPulse: Precision Biometric Sensing".
- Scientific Baccalaureate**

2021 – 2023

Joviat

Manresa
 - Graduated with Honors.
 - Research Project: "Automated Image Classification Pipeline".

EXPERIENCE

- Research Collaborator**

Feb 2025 – Mar 2025

ALBA Synchrotron

Cerdanyola, Spain
 - Contributed to research on **Barocaloric effects**.
 - Conducted high-precision data collection and measurements.
- Intern**

Jun 2022 – Aug 2022

Renewable Energy Company

Girona, Spain
 - Supported technical operations and data analysis.

TECHNICAL PROJECTS

- Modeling of the internal Tuberculosis Dynamics** | *Systems Biology*
 - Applied **Lanchester's Laws** to model the immune system's interaction with *M. tuberculosis*, analyzing population competition through differential equations.
- Mathematical modeling of Ardenas and Plennor Battles** | *Mathematical Physics*
 - Developed predictive models for the Battle of the Bulge and the Pelennor Fields, simulating attrition rates and tactical outcomes using Python-based numerical solvers.
- Optical Coherence Tomography (OCT) System** | *Optics, Signal Processing*
 - Developed and calibrated an OCT setup for non-invasive interferometric imaging.
- Laser Systems & Range Finding** | *Nd:YAG, Photonics*
 - Characterized Nd:YAG laser cavities and implemented a Laser Range Finder for precision distance measurement.
- Fluid Physics in Microgravity** | *Fluid Dynamics, Research*
 - Experimental study on fluid behavior and capillary effects under microgravity conditions.
- Thermodynamic & Crystallographic Analysis** | *Materials Science*
 - Conducted dielectric and structural characterization of disordered materials and phase-change compounds.
- Autonomous Line Tracker Robot** | *Control Theory, Electronics*
 - Designed and programmed a PID-controlled robot for high-speed trajectory tracking.